# Teaching sand to detect particles

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September 30th 2026

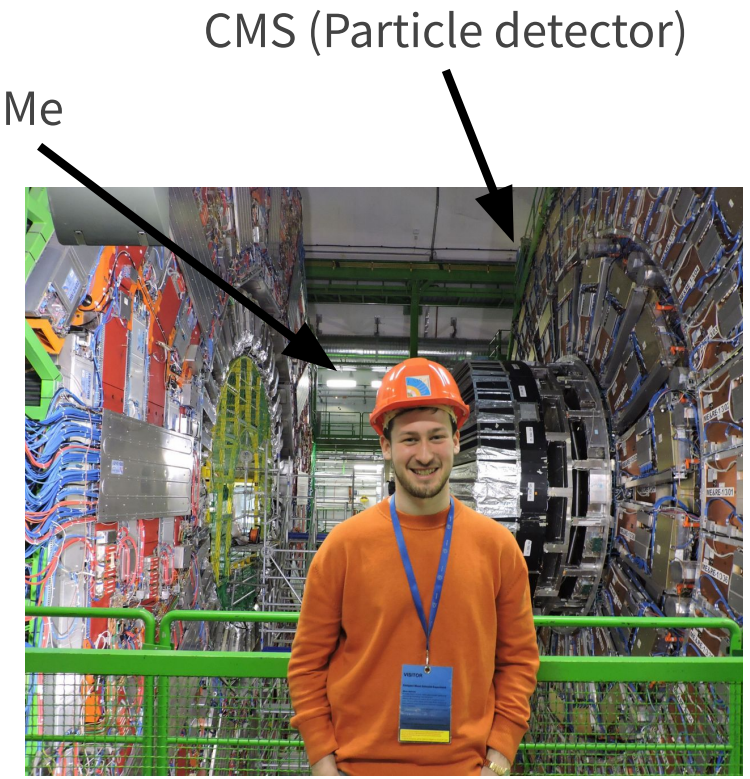


Brookhaven
National Laboratory



# How did I get here?

- **2012-2016: Undergrad** at Stony Brook University (B.S. in Physics)
- **2016-2022: PhD** in experimental particle physics at Northeastern University



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- **2022-now: Postdoc** at Brookhaven National Laboratory
- I work on **Higgs boson physics** and **tracking detectors**

ATLAS (Another particle detector)

Me



# Plan for this talk

- Introduce **particle physics**, walk through how we use **silicon particle detectors** to perform research.
- **Stop** in several places to see if there are questions, to **keep everyone on the same page**
- Give you an **introductory idea** of these topics.



# Plan for this talk

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- **Sto** are **th**
- Give you an **introductory idea** of these topics.

**Please raise your hand if you have a question!**



# Outline

Part I: Why and how we study particle physics

Part II: Detecting particles with silicon

# Next section

Brookhaven National Laboratory

## Part I: Why and how we study particle physics

Part II: Detecting particles with silicon

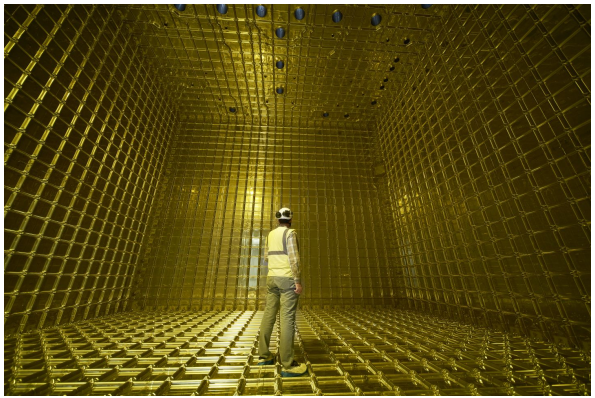
# Overview

**Why do we study particle physics?**

→ To understand nature at its most fundamental level: What is the universe made of? How do its constituents interact?

**How do we study particle physics?**

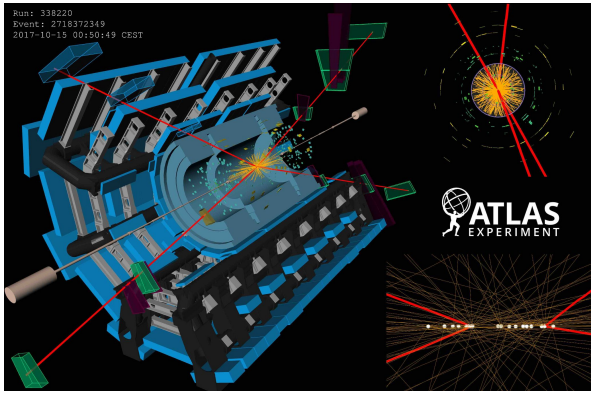
Intensity frontier



Cosmic frontier



Energy frontier



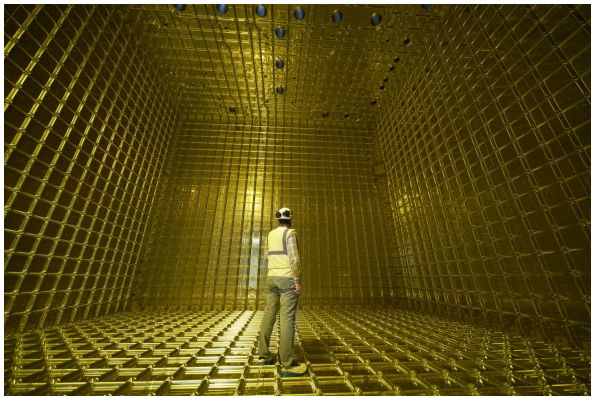
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**How do we study particle physics?**

Intensity frontier

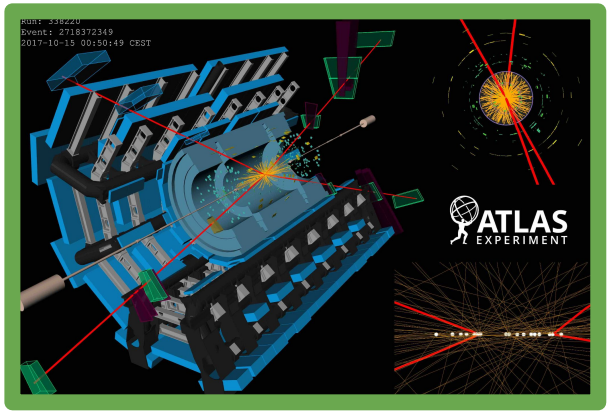


Cosmic frontier



Focus today:

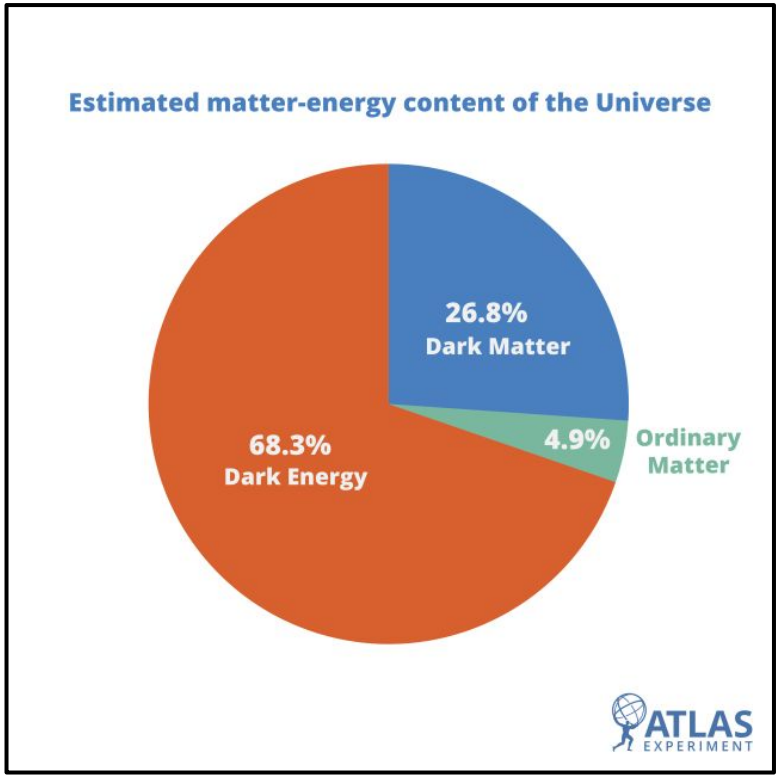
Energy frontier



# (Some of) The big physics questions

- Many deep, open questions:
  - Why is there **more matter** than **antimatter**?
  - What is the nature of **dark matter**?
  - How do **neutrinos** obtain mass?
  - …

- Different questions tied to different **particles we can study**
  - **For example:** Many of these questions have potential ties to the **Higgs boson**!



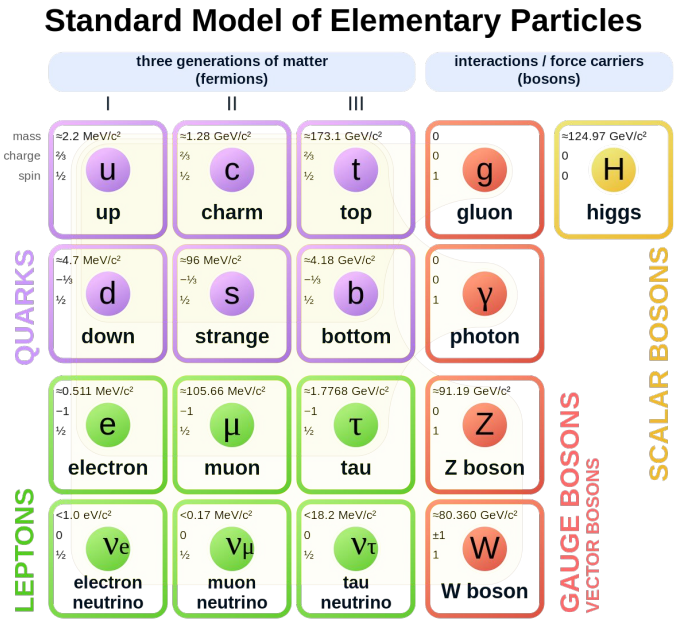

Searching for Dark Matter with the ATLAS detector

# The standard model

- **Extremely successful!** Predicts **vast majority** of observations

- **Does not explain everything:**
  - Origin of neutrino mass
  - Nature of dark matter
  - Matter-antimatter asymmetry
  - …

- Suggests the existence of **BSM (Beyond the Standard Model) physics**

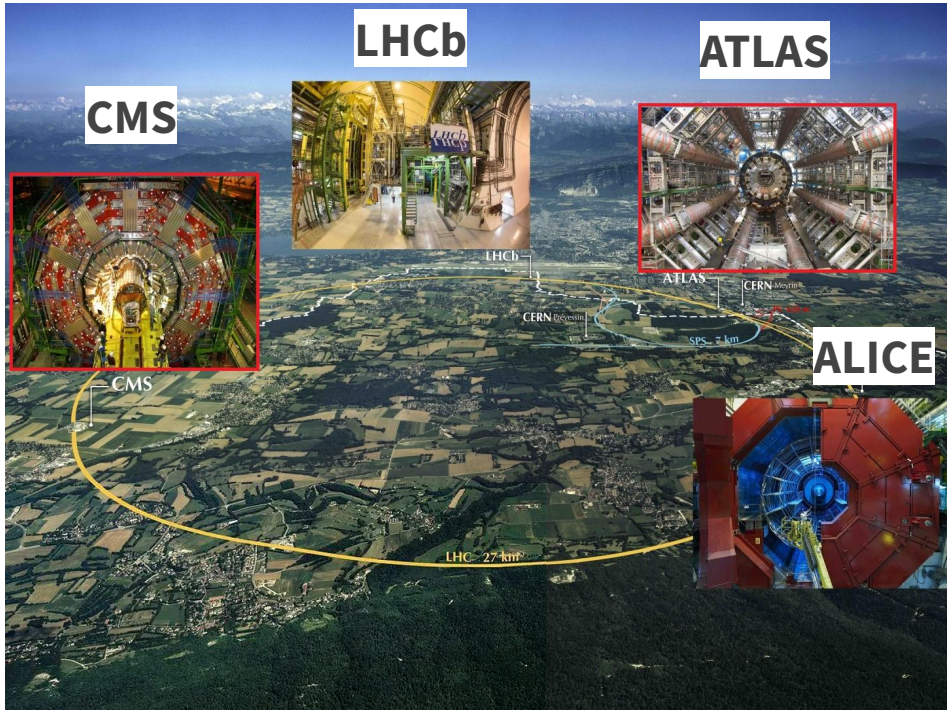




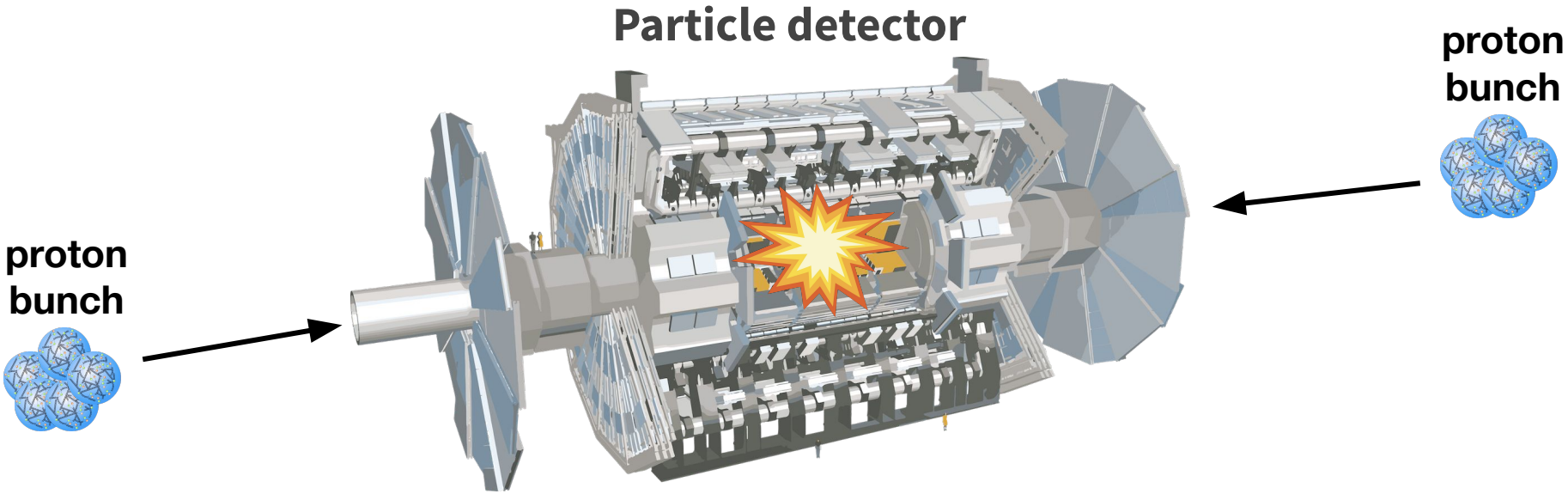

# The LHC

- To try answering our questions, need to **produce these particles in abundance**

- Collides **protons**, heavy ions up to **~ 99.999999%** the speed of light!
- Has **four detectors** stationed
- Operating since **2009**, will operate until **2041!**
- The LHC is a **Large** Hadron Collider:
  - **~ 20 minute drive from ATLAS to CMS!**



# The ATLAS detector




**Question:** **Why do we collide "bunches" of protons? Why not just collide 2 protons?**

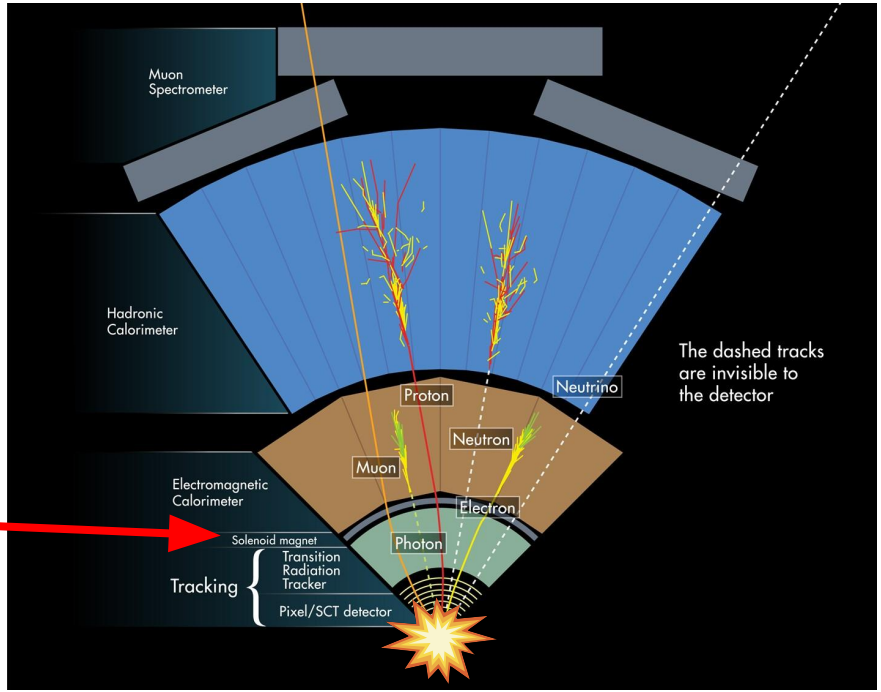
**Answer:** Very low chance of interaction with two protons. Collide **bunches of protons (~100 billion each!)**
**Pro:** **More likely to produce our physics processes of interest**
**Con:** **Have to deal with lots of extra interactions, making environment "messy".**

# The ATLAS detector

- Different **layers** detect different **particles**
- Different particles leave **different signatures**
- **Reconstruct collision** by working backwards from detector information

**Question:** Why does ATLAS have magnets?

**Answer:** To bend charged particles, use **Lorentz Force law** to measure **momentum**.



# Part I summary

- We study **particle physics** to answer **fundamental questions** about the universe.
- Particle **accelerators** are used to create **collisions**, particle **detectors** are used to **study what comes out of them**.
- We use different particle detector **technologies** to study different **particles**.

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**Any questions before we move to Part II?**



# Next section

Brookhaven National Laboratory

Part I: Why and how we study particle physics

**Part II: Detecting particles with silicon**

# Silicon

- Element, atomic number **14**
- Common in the **Earth's crust**, **rocks**, even **beach sand**! (Through $SiO_2$)
- Can **refine** $SiO_2$ to create more pure **Silicon**
- Defined as a "**metalloid**": Has traits of both **insulators** and **conductors**.




periodictable.com

# Processing Silicon



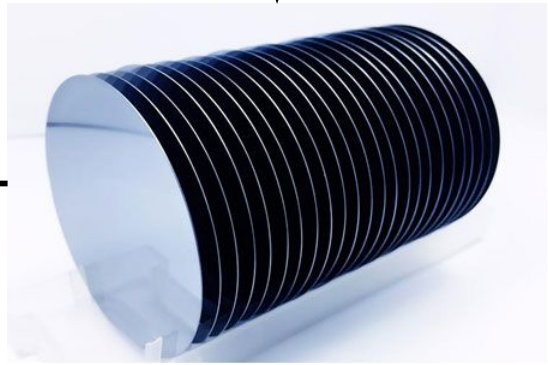

Refine



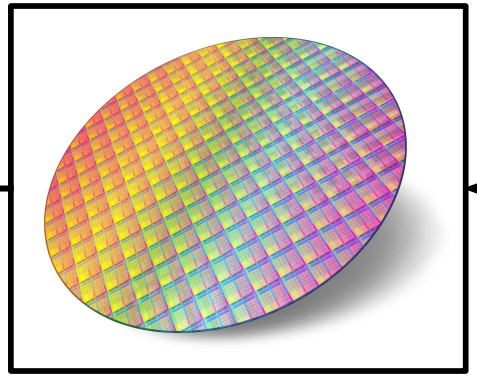
Melt + form



Slice



Etch the design

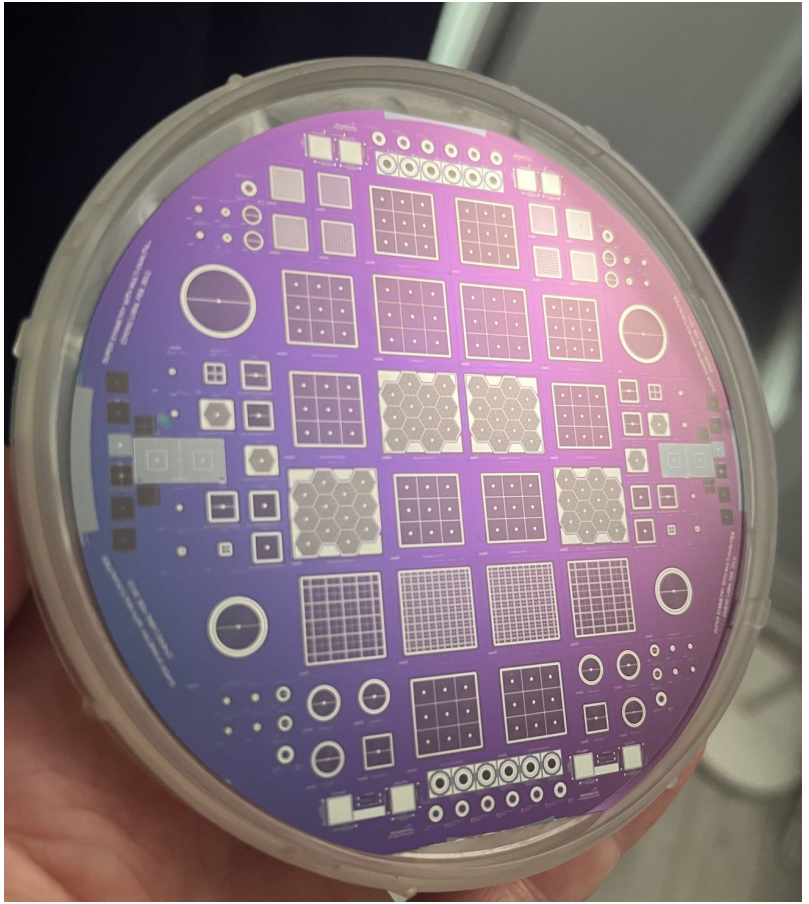


Final product!



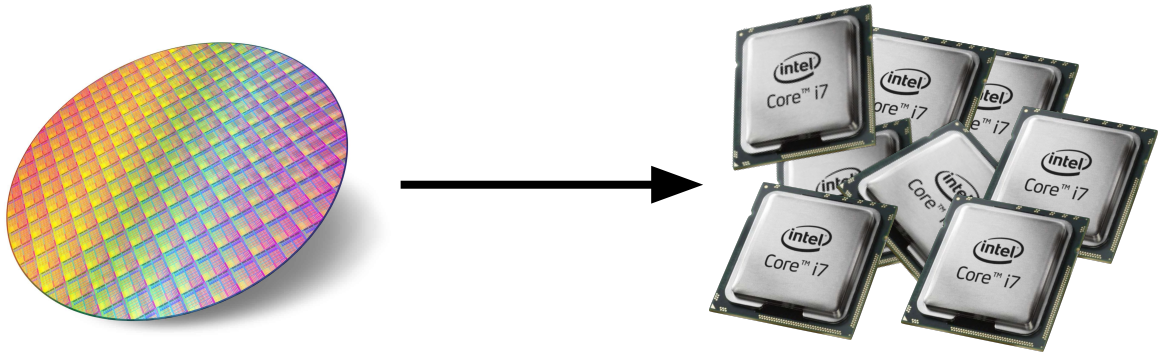
# Processing Silicon

Example wafers from BNL!



# Applications

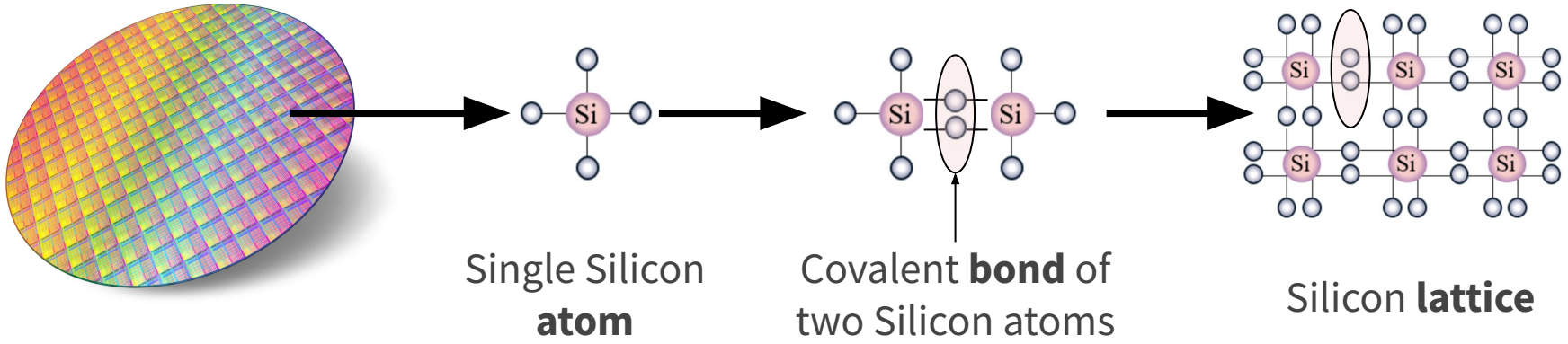
- Silicon is a staple in many **industrial** and **scientific** applications, e.g. **microelectronics**
- Widely used for **computer chips**, for example Central processing units (**CPUs**) and Graphic processing units (**GPUs**):




- Very deep experience in **industry** and **academia** in how to **process silicon for custom applications**.
- **How can we use silicon for particle physics?**

# The lattice

- Zooming into the completed silicon wafer
- Atomic number 14, so **4 valence electrons**




Single Silicon **atom**

Covalent **bond** of two Silicon atoms

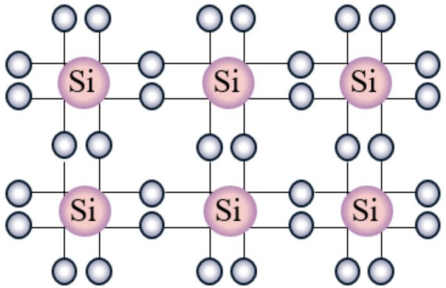
Silicon **lattice**

shindengen.com

# Doping

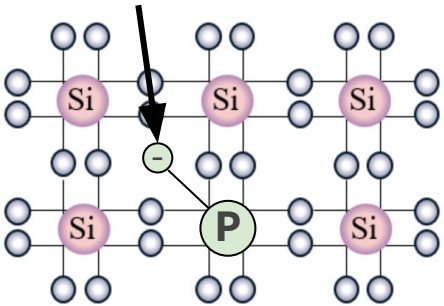
- On its own, a silicon lattice not particularly useful for physics - **does not conduct electricity!** (just have a bunch of **stable covalent bonds**)

- Produce **intentional impurities** called “**dopants**”:
  - **Replace silicon atoms in the lattice with other atoms**



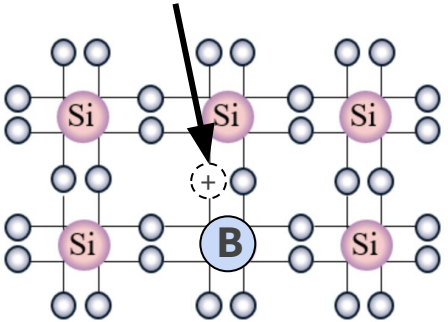

Silicon **lattice**

**Extra, free electron**




“**n-type**” doping

**Hole: Absence of electron**

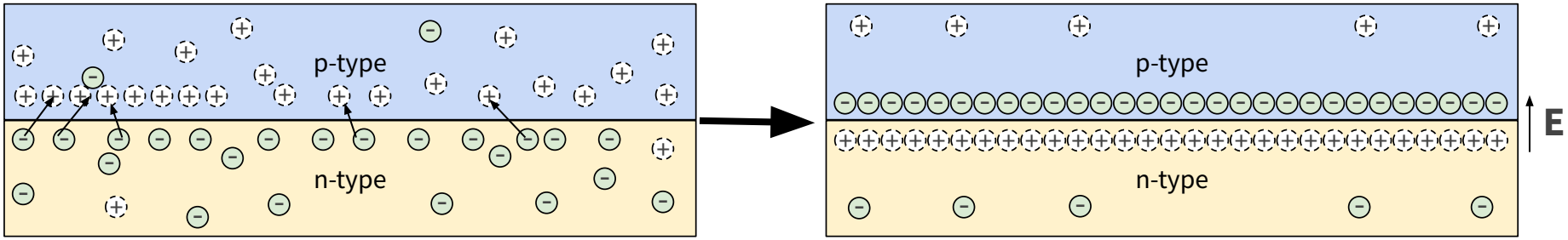



“**p-type**” doping

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# PN junction

- When p and n-type placed together, **diffusion** occurs, have electrons and holes naturally form boundaries near each other:



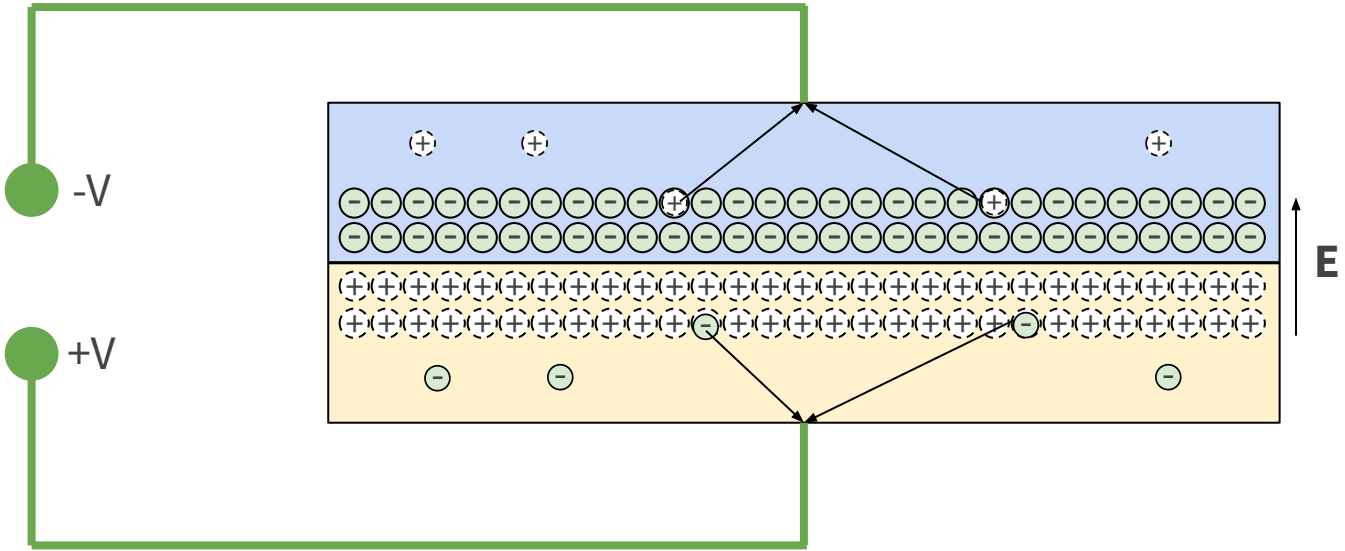

Effectively negative and positive charges.

**Question:** Any sort of **"field"** created between them?
**Answer:** Yes, an **electric field** (Coulomb's law - electric field from point charges)

→ Prevents further movement of electrons or holes across the boundary - call this the **depletion region** (no "free" electrons or holes)

# Bias voltage

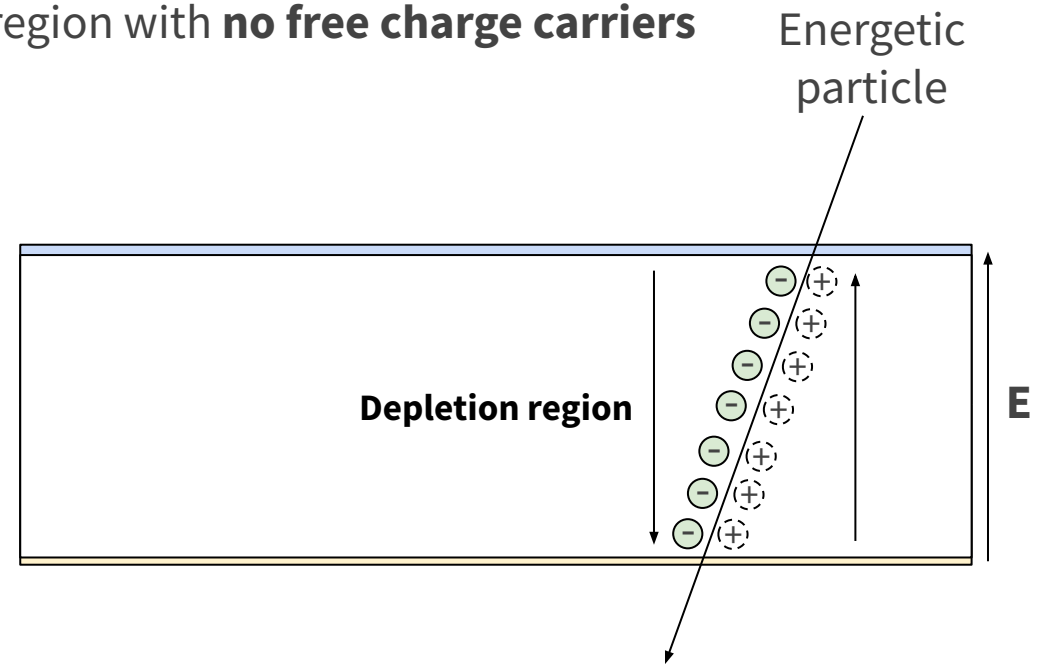
- What happens if we apply a **voltage** across the PN junction?




- Remove holes from top, leaves negative charge behind (-).
- Remove electrons from bottom, increase net charge in its place (+).
- **Increase in the depletion region**

# Ionization and collection

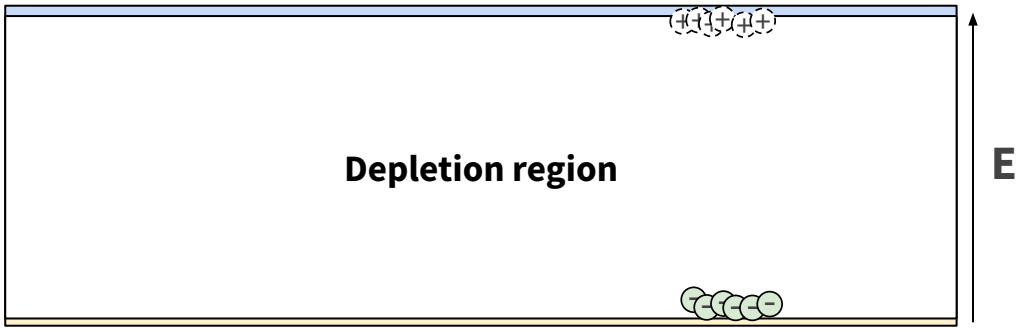
- Now have a region with **no free charge carriers**




**Question:** What will happen to these new **free** electrons and holes?
**Answer:** They will experience an **electric force** and accelerate in either direction

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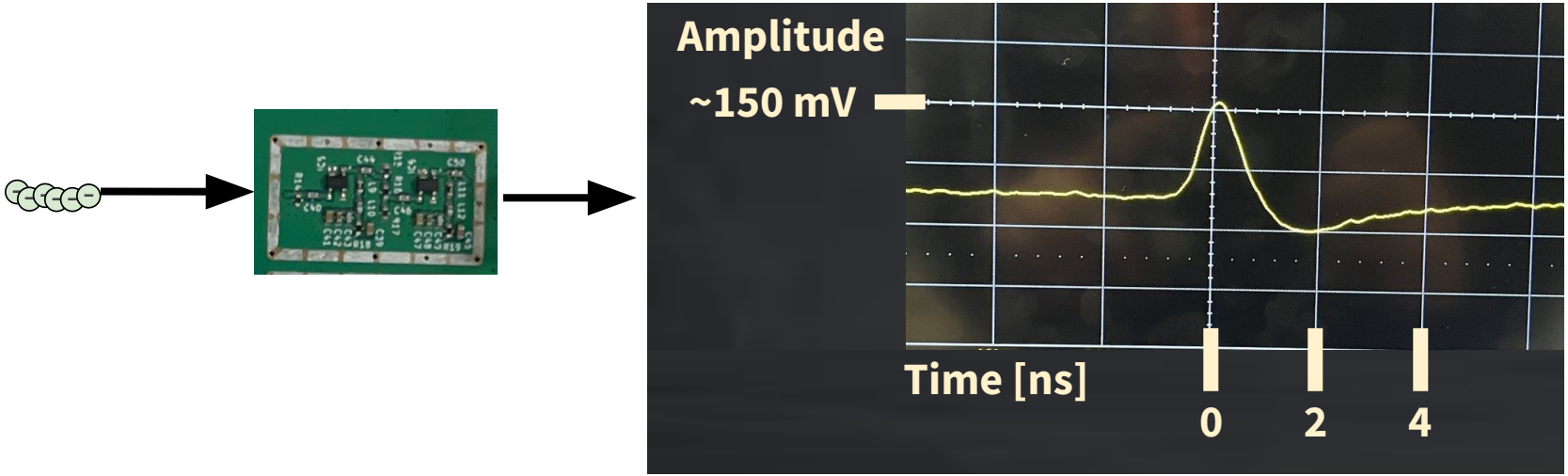



**Question:** What will happen to these new **free** electrons and holes?
**Answer:** They will experience an **electric force** and accelerate in either direction

This means we can **collect charge** on either side of the device.

# Waveforms

- When the charges move, this creates **electric current** (charge as a function of time)
- Use **electronics** to **amplify** and **shape** this current
- Gives us a **waveform** (example I took at **BNL**):

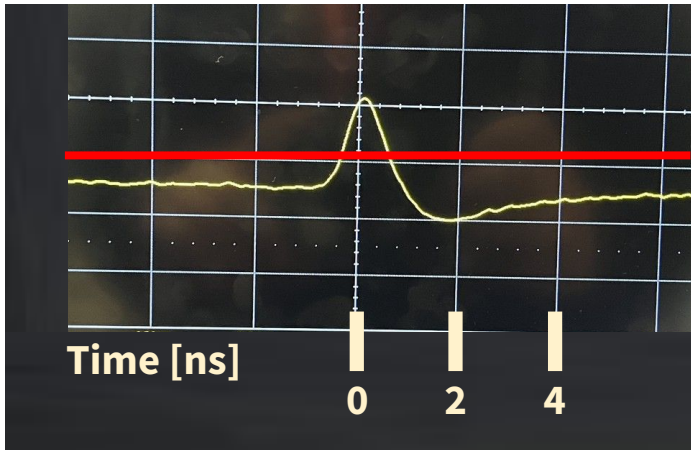



Signal lasts for **~ a billionth of a second**!!

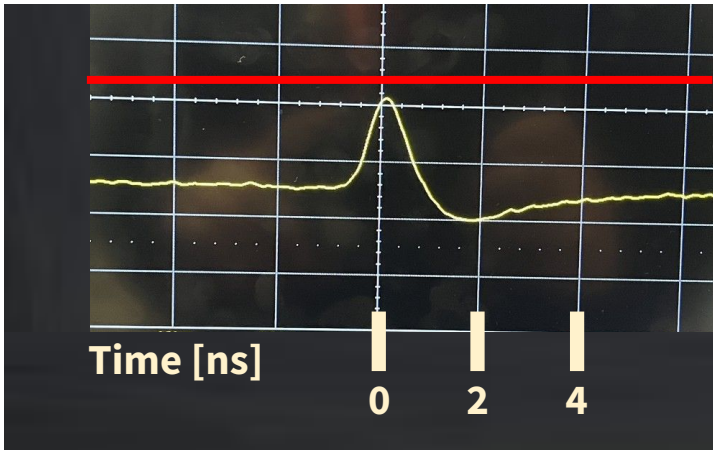
- This means if we see a **waveform**, it might be a **signal from a particle**!

# Signal analysis

- **What can we do with this signal waveform?**
- One example: **Binary classification**
- If waveform goes above a certain height, consider this a **hit**




Classify as a **hit**




Classify as **not a hit**

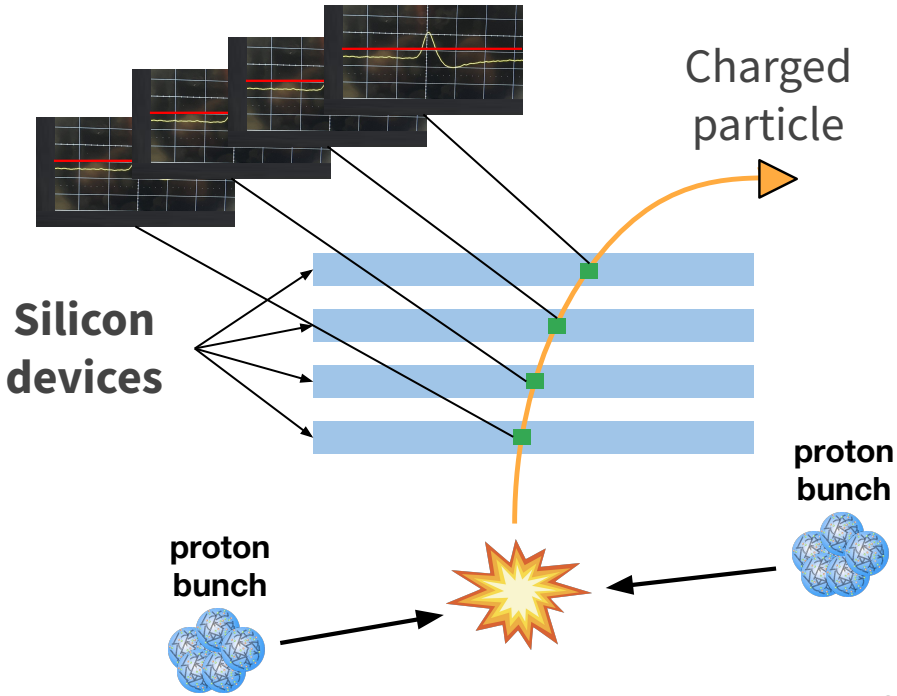
- **Question:** Why don't we just set the height **very low** so that everything we see is a hit?
- **Answer:** Want to avoid **noise** (very common from **silicon** and **electronics**).

# Example

- **Question:** How can this be useful for particle physics?
- **Answer:** Tell us if we detected a particle from a collision
- Example: Track charged particles!

- Use these measurements to do lots of things e.g.:
  - Measure **momentum** of charged particles
  - Look for **new particles**

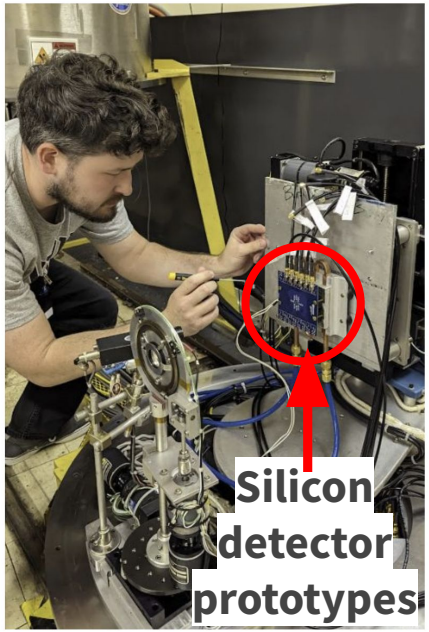
- All with the goal of answering our **original physics questions**!



# Research

- There is a lot of **very cool research** ongoing for silicon detectors!






**Something even better than silicon (?)**

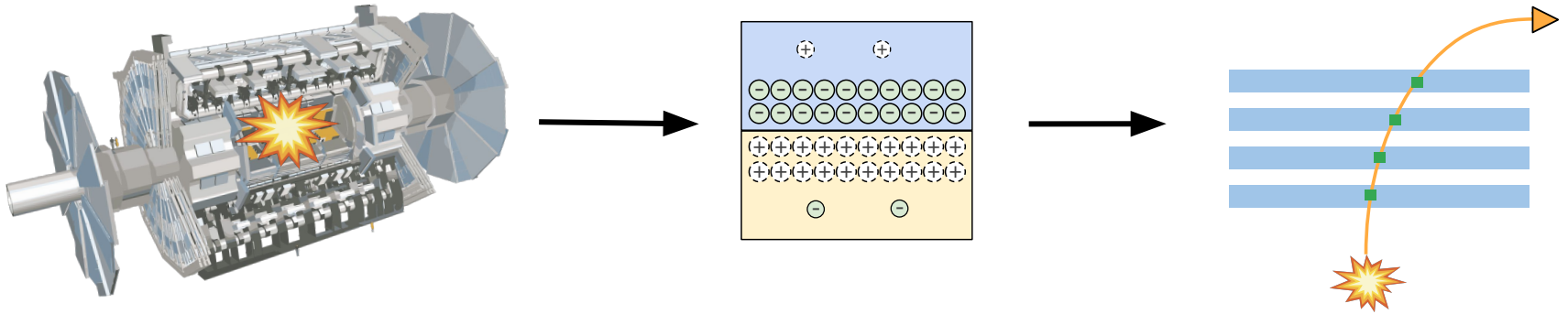


# Summary



# Summary and main takeaways

- **Silicon** is a widespread technology in **industry** and **academia**
- For **particle physics**, we use silicon to **detect charged particle** from **particle collisions**
- Use measurements to try and answer **fundamental questions about the universe**
- Lots of **active research** in silicon detectors. **Very fun place to get involved**!
- Feel free to **reach out** to me if you have questions in the future!
  → ***abraham.tishelman.charny@cern.ch***




Thank you for your attention!